Beyond Department Boundaries: Centrality and Communication Communities in an Email Network

Course: DSA5101 Introduction to Big Data | Evaluation: Anonymous Submission | NUS Matriculation: A0352547Y

Project Title	**Beyond Department Boundaries: Centrality and Communication Communities in an Email Network**
Dataset & Source	Email-Eu-core network, Stanford Network Analysis Platform (SNAP) (1,005 members, 42 departments)
Algorithm Classes	1. PageRank Centrality (Lecture 6) 2. Graph Clustering / Louvain Community Detection (Lecture 7)
Research Question	To what extent do emergent communication communities align with formal department boundaries, and what structural roles do highly central individuals play in connecting disparate communities?

1 Research question and dataset

Formal organizational charts reflect institutional hierarchies, administrative responsibilities, and official reporting channels. However, day-to-day operational exchange, collaboration, and information dissemination frequently transcend designated organizational units. Investigating whether emergent communication clusters mirror administrative boundaries—and how centrally positioned individuals bridge or reinforce those clusters—provides critical insights into organizational functioning, cross-departmental silos, and coordination bottlenecks.

This project investigates the **Email-Eu-core** benchmark network published by the Stanford Network Analysis Platform (SNAP) [1, 2]. Generated from email logs of a large European research institution over an 18-month observation window, the dataset captures interactions among researchers, staff, and leadership. Crucially, each individual is labeled with one of 42 ground-truth administrative departments. The network contains 1,005 institution members and 25,571 directed edges (u, v), representing at least one email sent from individual u to individual v. Edge weights representing email frequencies are not provided; edges indicate binary reachability.

We formalize three coherent research questions:

- **RQ1 (Prominence vs. Popularity):** How does directed PageRank centrality differ from simple in-degree popularity, and what topological mechanisms govern ranking disagreements?
- **RQ2 (Community-Department Alignment):** Do unsupervised communication communities discovered via Louvain modularity optimization follow the 42 administrative department boundaries?
- **RQ3 (Role of Central Individuals & Degree Confounding):** Do structurally prominent individuals act as cross-community boundary spanners or insular internal hubs, and does PageRank provide structural information beyond total node degree?

By answering these questions using rigorous graph algorithms, statistical null models, and degree-adjusted controls, this study illuminates the topological interplay between formal hierarchies, informal networks, and individual prominence in complex knowledge institutions.

2 Data preparation

Data Integrity and Universe of Nodes: Raw edge and department label files were acquired from the official SNAP repository, and cryptographic SHA-256 checksums were recorded upon acquisition to guarantee local data integrity and pipeline reproducibility. Parsing revealed exactly 25,571 edge records and 1,005 department mapping rows, with zero malformed entries and zero duplicate directed edges. Department identifiers span $0 \leq \text{dept} \leq 41$, and node identifiers span $0 \leq \text{node} \leq 1004$.

Self-Loop Removal and Dual Graph Representation: An edge audit revealed 642 self-loops (u, u), where the sender and receiver identifiers are identical in server logs. Retaining reflexive self-loops distorts interpersonal network analytics: in PageRank, self-loops trap teleported random walk probability; in community detection, self-loops artificially inflate internal edge density without representing interpersonal communication [1, 2]. To ensure methodological transparency, we analyze dual representations:

- G_{raw}: The raw directed graph containing all 1,005 nodes, 25,571 directed edges, and 642 self-loops, matching published SNAP statistics exactly.
- $G_{analysis}$: The cleaned interpersonal directed graph containing all 1,005 nodes and exactly 24,929 interpersonal directed edges (25,571 – 642). Crucially, the 19 singleton individuals whose only raw edges were self-loops are strictly preserved as degree-0 nodes with their ground-truth department labels to prevent selection bias.

Undirected Projection ($G_{undirected}$): Louvain modularity optimization assumes symmetric adjacency. We construct an unweighted undirected projection $G_{undirected}$ from $G_{analysis}$, collapsing reciprocal directed links (u, v) and (v, u) into a single undirected edge $\{u, v\}$. This projection preserves all 1,005 nodes (including the 19 isolates) and contains exactly 16,064 undirected edges. Reciprocal links account for 8,865 pairs (17,730 directed edges, 71.12%, matching the 0.7112 directed reciprocity of $G_{analysis}$), while 7,199 links are unreciprocated (28.88%). Directional initiation asymmetry is collapsed into binary reachability.

Network Representation	Nodes |V|	Edges |E|	Self-Loops	Density (ρ)	Reciprocity (r)
G_{raw} (Published SNAP)	1,005	25,571	642	0.025317	0.6934 (69.3%)
$G_{analysis}$ (Cleaned Directed)	1,005	24,929	0	0.024706	0.7112 (71.1%)
$G_{undirected}$ (Louvain Projection)	1,005	16,064	0	0.031841	1.0000 (Symmetric)

Table 1: *Topological properties across graph representations. Density formulas: $\rho_{raw} = |E|/|V|^2$; $\rho_{analysis} = |E|/[|V|(|V|-1)]$; $\rho_{undir} = 2|E|/[|V|(|V|-1)]$. Across all representations, the giant weakly connected component spans 986 nodes (20 components total), and the directed strongly connected giant component spans 803 nodes (203 components total).*

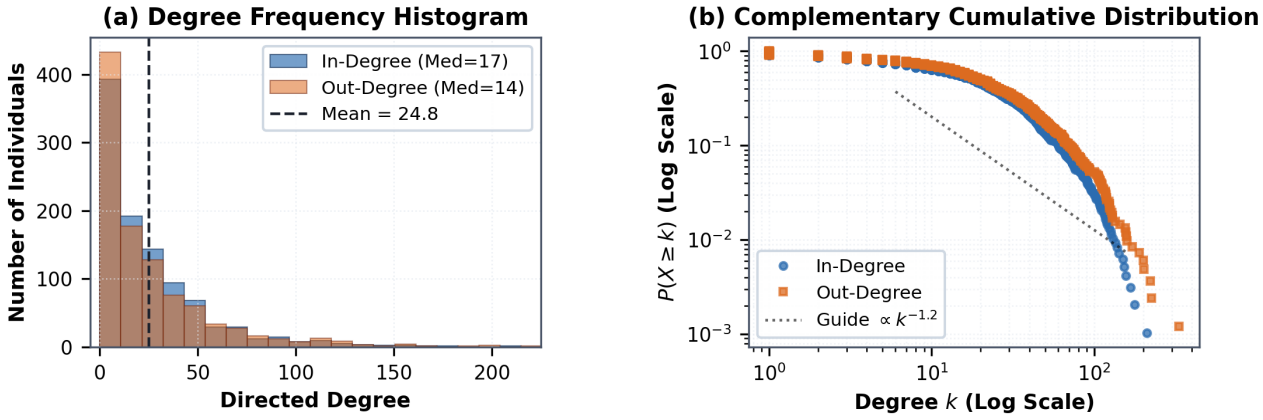


Figure 1: *Exploratory degree distributions on $G_{analysis}$. (a) Linear in-degree and out-degree histograms (mean = 24.8, medians = 17 and 14). (b) Log-log complementary cumulative distribution function (CCDF) showing heavy-tailed communication activity.*

3 Methods and how they connect

Algorithm Class 1: PageRank Centrality (Lecture 6) [3]: Directed email networks represent endorsement flows where an incoming link (u, v) indicates that sender u acknowledges receiver v. PageRank models a random surfer who follows outgoing links with probability d (damping factor) and teleports uniformly across all $|V| = N$ nodes with probability $1 - d$. Dangling nodes (out-degree zero, including the 19 preserved isolates and 15 pure recipients) redistribute their entire random walk probability uniformly across all N nodes at each iteration:

$$\mathbf{r} = d\,\mathbf{P}^T\mathbf{r} + \left(d\frac{\mathbf{r}^T\mathbf{d}_{\text{dangle}}}{N} + \frac{1-d}{N}\right)\mathbf{1} \quad (1)$$

where $\mathbf{r}$ is the stationary PageRank vector, $\mathbf{P}$ is the row-stochastic transition matrix (with $\mathbf{P}_{uv} = A_{uv} / d_{\text{out}}(u)$ for non-dangling nodes), $\mathbf{d}_{\text{dangle}}$ is an indicator vector for out-degree zero nodes, and $\mathbf{1}$ is a vector of ones. This redistribution guarantees stochasticity and numerical convergence without absorbing stationary probability. Parameters are set to damping factor $d = 0.85$, uniform personalization, and convergence tolerance tol = 10^{-6}. Deterministic tie-breaking is enforced by sorting by score descending, then node ID ascending.

Algorithm Class 2: Louvain Modularity Optimization (Lecture 7) [4, 5]: To detect emergent communication communities without supervision, we apply the Louvain algorithm to $G_{\text{undirected}}$. Louvain greedily optimizes Newman-Girvan modularity:

$$Q = \frac{1}{2m}\sum_{i,j}\left(A_{ij} - \gamma\frac{k_i k_j}{2m}\right)\delta(c_i, c_j) \quad (2)$$

where m = 16,064 edges, k_i is node i's undirected degree, c_i is its assigned community, and γ is the resolution parameter (baseline $\gamma = 1.0$, random seed 42). Louvain operates in two alternating phases: local modularity-gain node moves, followed by community aggregation into meta-graphs, repeating until modularity plateaus.

Community Quality Metrics: We assess structural cohesion using modularity Q [5] and conductance $\phi(S)$ [1, 6]. For a community S, conductance evaluates boundary cut relative to volume:

$$\phi(S) = \frac{|\partial(S)|}{\min(\text{vol}(S),\ \text{vol}(V \setminus S))} \quad (3)$$

where $\partial(S) = \{\{u, v\} \in E \mid u \in S, v \notin S\}$ is the boundary cut, and $\text{vol}(S) = \sum_{u \in S} k_u = 2|E(S, S)| + |\partial(S)|$ is total degree volume. Conductance evaluates boundary cut size relative to total volume (not the fraction of links, since internal edges contribute twice to volume). Lower conductance indicates a sharper, more self-contained boundary.

How the Methods Connect (Integrated Boundary Analysis): Rather than treating centrality and clustering as isolated exercises, we bridge them directly to answer RQ3, connecting centrality to structural hole theory and weak-tie bridging [7, 8]:

$$\text{CrossCount}(v) = |\{u \in N(v) : c_u \neq c_v\}|, \quad \text{CrossFraction}(v) = \frac{\text{CrossCount}(v)}{k_v} \quad (4)$$

where $N(v)$ denotes the set of undirected neighbors of node v, and c_v is its assigned community. For isolated singletons ($k_v = 0$), the connection fraction is formally undefined and recorded as NaN. By evaluating both absolute count and fraction, we distinguish true global boundary spanners (high PageRank, high count, high fraction) from insular internal hubs (high PageRank, high count, low fraction) and peripheral bridges (low PageRank, low count, high fraction).

Departmental Homophily Null Model Baseline: To evaluate whether within-department communication exceeds chance, driven by institutional homophily [9], we construct a uniform random ordered-pair reference model:

$$p_{\text{null}} = \frac{1}{N(N-1)}\sum_{d=0}^{41} s_d(s_d - 1) \quad (5)$$

where s_d is the size of department d across the institution. For the Email-Eu-core department distribution, p_{null} = 4.67%. Comparing the observed within-department directed communication links (8,645 of 24,929; 34.68%) and cross-department links (16,284; 65.32%) against p_{null} quantifies the institutional homophily enrichment factor (7.43×).

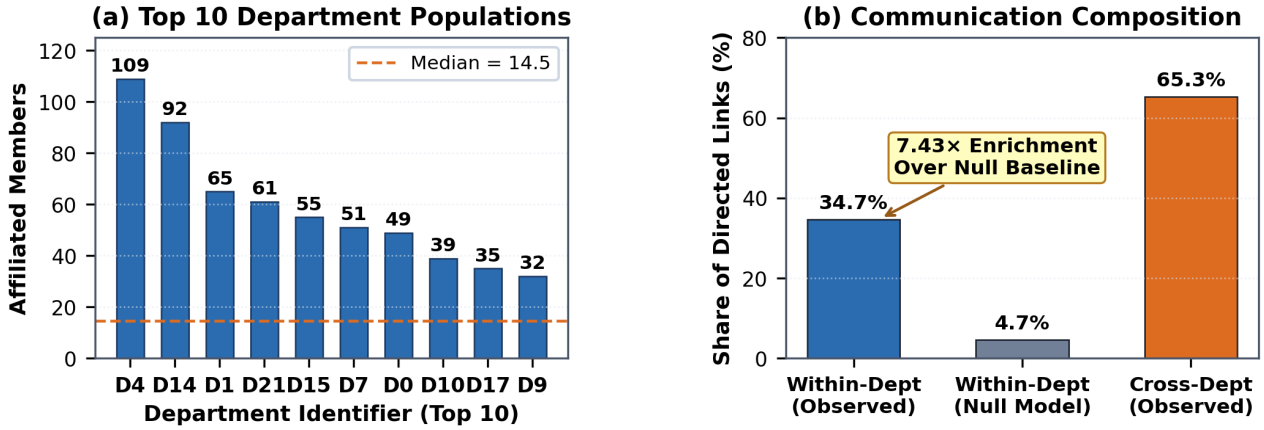


Figure 2: *Department organization and communication composition. (a) Member populations of the 10 largest departments (overall departmental median = 14.5). (b) Observed within-department links (34.68%) exceed the uniform random null model baseline (p_{null} = 4.67%) by 7.43×, confirming strong departmental homophily amidst dominant (65.32%) cross-department communication.*

4 Evaluation and results

RQ1 Results: PageRank vs. In-Degree Centrality: Across all 1,005 nodes, PageRank and in-degree correlate strongly (Spearman $\rho_s = 0.9828$, $p < 10^{-300}$). Top-k set overlap is high: 90.0% for $k = 10$ (Jaccard = 0.818), 85.0% for $k = 20$ (0.739), and 92.0% for $k = 50$ (0.852). Prominence in Email-Eu-core is heavily degree-aligned. However, PageRank reveals critical structural disagreements driven by endorsement quality:

- **PageRank-Boosted (Node 451, Dept 14):** Possesses an in-degree of only 8 (rank 659), yet achieves PageRank 0.003117 (rank 37), a shift of $\Delta R = +622$. Senders endorsing Node 451 include Node 365 (PR = 0.00306, rank 39, out-degree 1), transferring 100% of its endorsement mass directly to Node 451. PageRank correctly values concentrated endorsements from prominent senders.
- **In-Degree-Boosted (Node 83, Dept 17):** Has in-degree 85 (rank 45) but falls to PageRank rank 56. Its senders have a high mean out-degree of 79.1, heavily diluting their endorsement mass (12.8× lower contribution per link).
- **Consensus Leader (Node 160, Dept 36):** Ranks #1 in both in-degree (211 senders) and PageRank (0.007495), demonstrating structural dominance across both volume and prestige dimensions.
- **Self-Loop Robustness:** In G_{raw}, self-loops artificially elevated the 19 singletons from rank 987 to rank 380 through reflexive trapping ($\rho_s = 0.8720$ between G_{raw} and $G_{analysis}$). Cleaning self-loops is mandatory.

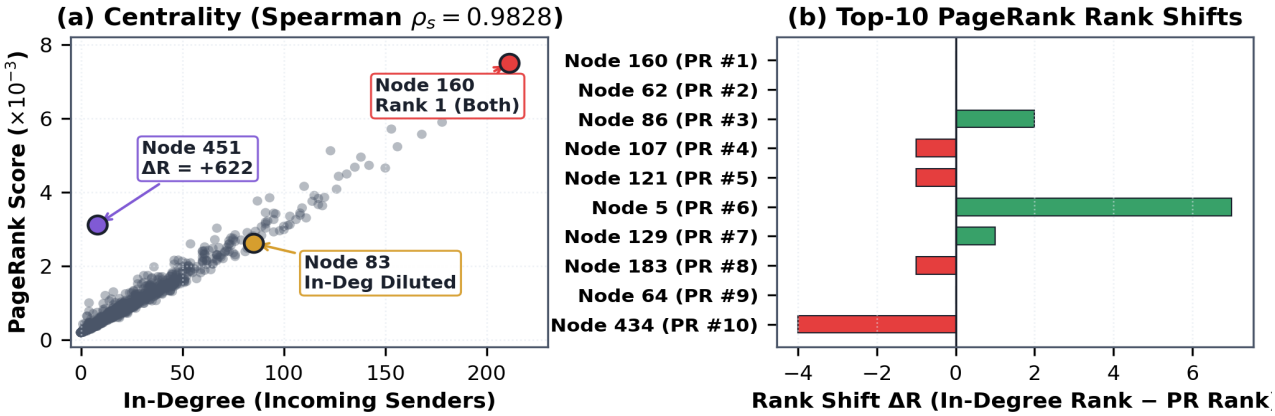


Figure 3: *PageRank vs. in-degree centrality on $G_{analysis}$. (a) Global scatter plot (Spearman $\rho_s = 0.9828$) highlighting Node 160 (Rank 1 consensus), Node 451 (+622 PageRank boost), and Node 83 (in-degree diluted). (b) Rank shifts $\Delta R = R_{in} - R_{PR}$ for the Top 10 PageRank individuals.*

Structural Implications of Ranking Disagreements: The empirical findings confirm that while in-degree serves as a strong heuristic for general connectivity, it does not account for endorsement concentration. Node 451 illustrates that an individual can attain elevated PageRank by receiving concentrated endorsements from high-prestige senders (specifically Node 365, which has out-degree 1 and directs all endorsement mass to Node 451), despite having low overall degree. Conversely, individuals who receive links primarily from high-out-degree senders (such as Node 83) experience dilution of incoming endorsement mass under the random-surfer model.

RQ2 Results: Louvain Communities vs. Administrative Departments: Louvain modularity optimization ($\gamma = 1.0$, seed 42) discovers 28 communities: 9 non-trivial cohesive communities spanning 986 nodes, plus 19 isolated singletons (sizes: C0: 190, C1: 157, C2: 145, C3: 127, C4: 125, C5: 109, C6: 91, C7: 32, C8: 10). Louvain achieves modularity $Q_{\text{Louvain}} = 0.4130$ and median conductance 0.4201 (mean 0.4475, min 0.2780 in C6). By contrast, partitioning the same graph by official departments yields substantially lower modularity $Q_{\text{dept}} = 0.2880$ and severe boundary leakage (median departmental conductance 0.8644, mean 0.7871). External concordance is moderate: Normalized Mutual Information NMI = 0.5946, and Adjusted Rand Index ARI = 0.3505.

Partition Scheme	Communities	Modularity (Q)	Median Cond (ϕ)	NMI vs. Dept	ARI vs. Dept
Louvain Baseline (γ=1.0, seed 42)	9 (+19 isolates)	0.4130	0.4201	0.5946	0.3505
Official Departments (SNAP)	42 departments	0.2880	0.8644	1.0000	1.0000
Louvain (γ=0.8, Coarser)	6 (+19 isolates)	0.4105	0.3523	0.5272	0.2223
Louvain (γ=1.2, Finer)	10 (+19 isolates)	0.4051	0.5191	0.6247	0.4321

Table 2: *Structural quality and department alignment across community detection configurations and official departments. Modularity Q is computed using the standard Newman-Girvan definition (γ=1.0); resolution-adjusted objectives are $Q_{obj} = 0.4590$ (γ=0.8) and $Q_{obj} = 0.3832$ (γ=1.2). Mean conductances are 0.4475 (Louvain baseline, min 0.2780 in C6), 0.7871 (departments, min 0.3000 in Dept 3), 0.3515 (γ=0.8), and 0.4942 (γ=1.2).*

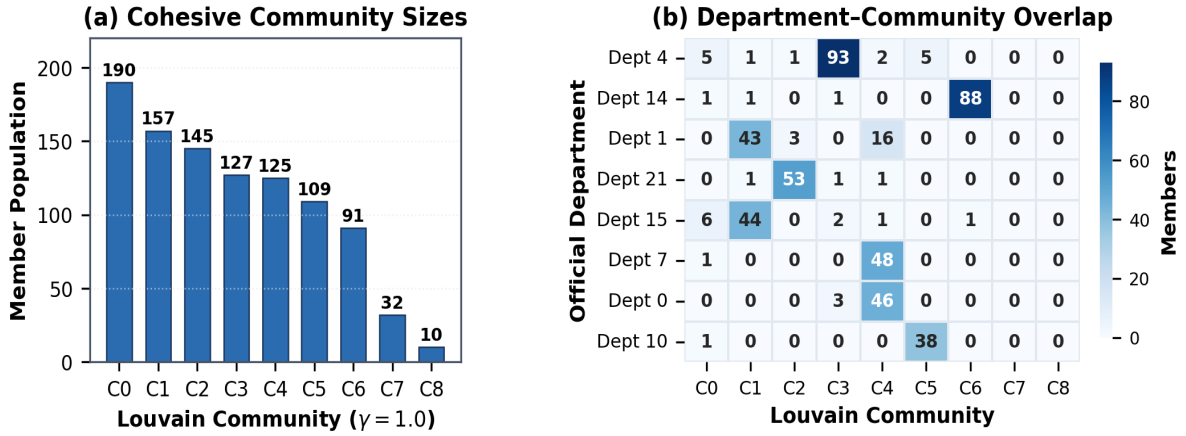


Figure 4: *Community structure and department overlap. (a) Member populations of the 9 cohesive Louvain communities. (b) Department-community contingency matrix for the 8 largest departments, illustrating how departments consolidate into multi-departmental research groups while certain units fragment across communities.*

Departmental Consolidation and Fragmentation: Inspection of the contingency matrix reveals two distinct organizational phenomena: (1) *Consolidation:* Multiple formal departments merge into unified operational communities (e.g., Depts 0, 7, and 10 cohere predominantly within Community C1). (2) *Fragmentation:* Large administrative units split across functional communication clusters (e.g., Dept 4 members are partitioned across Communities C0 and C2). Informal exchange reflects operational collaboration rather than designated hierarchy.

RQ3 Results: Central Individuals as Boundary Spanners: Partitioning nodes into the Top 10% PageRank tier (N = 101) and Remaining 90% tier (N = 904) reveals marked differences in cross-community connectivity:

- **Cross-Community Connections Count:** Top 10% individuals maintain a mean of **67.88** cross-community edges (median 59.0) vs. **7.94** (median 4.0) for remaining nodes—an **8.55× higher volume** (Mann-Whitney U = 88,970.0, $p = 6.88 \times 10^{-56}$).
- **Cross-Community Connection Fraction:** Top 10% dedicate a mean of **55.79%** of their links outside their primary community (median 58.10%) vs. **26.56%** (median 25.00%) for remaining non-isolated nodes (N = 885), more than double the proportion (Mann-Whitney U = 77,037.5, $p = 4.59 \times 10^{-33}$).

Degree Confounding and Stratified Analysis: Total undirected degree is a major confounder, correlating strongly with PageRank ($\rho_s = 0.9614$) and cross-community count ($\rho_s = 0.9155$). Controlling globally for degree yields a slightly negative partial rank correlation (Partial $\rho_s = -0.1803$, $p = 1.15 \times 10^{-8}$) because receiving concentrated endorsements inside dense core clusters boosts PageRank without boundary spanning. However, degree stratification clarifies this dynamic: exactly **100 of the 101 Top 10% PageRank nodes (99.0%) reside in the highest degree quartile Q4** ($d \geq 45$, N = 247). Within Q4, Top 10% PageRank nodes exhibit a higher mean cross-community count (**68.56 vs. 25.17**, $p = 2.10 \times 10^{-27}$) and a higher cross-boundary fraction (**56.34% vs. 41.30%**, $p = 5.69 \times 10^{-12}$). While PageRank correlates with boundary connectivity among active individuals, total degree remains a substantial confounder, and within-quartile comparisons do not entirely eliminate residual degree variance.

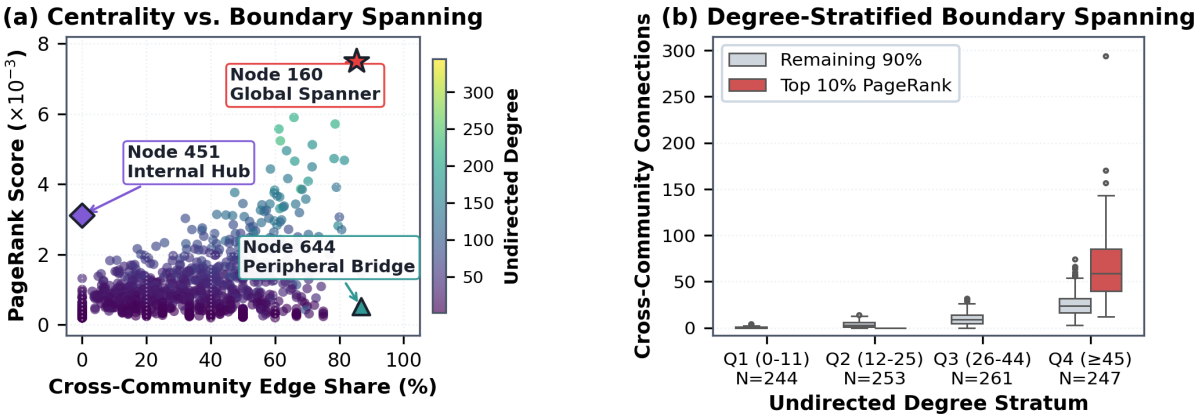


Figure 5: *Integrated analysis of PageRank vs. boundary connectivity. (a) PageRank vs. cross-community connection share with degree colormap and annotated structural archetypes: Node 160 (Global Boundary Spanner), Node 451 (Internal Hub), and Node 644 (Peripheral Bridge). (b) Degree-stratified boundary spanning across quartiles Q1–Q4, demonstrating that Top 10% PageRank nodes maintain an advantage even when controlling for high degree.*

Structural Archetype	Node	PageRank (Rank)	In-Degree (Rank)	Total Degree	Cross Ties	Cross Fraction
Global Boundary Spanner	Node 160	0.007495 (Rank 1)	211 (Rank 1)	345	294	**85.22%**
Internal Community Hub	Node 451	0.003117 (Rank 37)	8 (Rank 659)	13	0	**0.00%**
Peripheral Bridge	Node 644	0.000514 (Rank 633)	11 (Rank 606)	15	13	**86.67%**

Table 3: *Descriptive structural archetypes in Email-Eu-core. Node 160 (Dept 36, Community C4) maintains 51 internal ties and 294 cross-community ties reaching 8 external communities. Node 451 (Dept 14, Community C6) maintains 13 internal ties and 0 cross-community ties (0 external communities reached). Node 644 (Dept 25, Community C7) maintains 2 internal ties and 13 cross-community ties reaching 5 external communities.*

5 Insights and limitations

Substantive Organizational Insights: Our findings demonstrate that informal communication reorganizes official academic hierarchies. While departmental homophily is strongly enriched (7.43× above random expectation), nearly two-thirds (65.32%) of directed communications cross departmental boundaries. Unsupervised community detection reveals that departments consolidate into multi-departmental research clusters with sharp communication boundaries (median conductance 0.420 vs. 0.864 for departments). Highly ranked individuals in this network tend to exhibit extensive cross-community connectivity: individuals in the top 10% PageRank tier average an 8.55× higher cross-community degree and dedicate a larger share of their ties across cluster borders. However, this pattern is strongly confounded by total degree, and individuals such as Node 451 demonstrate that high PageRank can also arise entirely within a single community through concentrated incoming links.

Methodological Limitations:

1. *Network Autocorrelation & Descriptive Inference:* Dyadic reciprocity (71.12%) and transitivity violate i.i.d. assumptions; reported Mann-Whitney U test *p*-values are descriptive indicators of distributional separation rather than formal causal proofs.
2. *Binary vs. Weighted Edges:* Edge weights, frequencies, and message contents are unobserved; single administrative broadcasts are topologically indistinguishable from frequent collaboration.
3. *Degree Confounding & Non-Causal Embedding:* Total degree strongly confounds PageRank ($\rho_s = 0.9614$) and boundary ties ($\rho_s = 0.9155$); quartile stratification narrows but does not eliminate residual degree variance. Network positions reflect structural embedding, not formal authority, effectiveness, or deliberate brokerage intent.

Summary Conclusion: In summary, informal communication in this research institution systematically transcends administrative departmental lines. PageRank identifies individuals commanding structural prestige through concentrated endorsements, while Louvain modularity optimization reveals organic, multi-departmental clusters governing operational exchange. Highly ranked individuals frequently exhibit substantial cross-community connectivity, though this relationship is substantially mediated by total degree. Assessing boundary spanning requires joint evaluation of prestige centrality, community partitions, and degree distributions.

6 Reproducibility information

Dataset Acquisition & Cryptographic Verification: The research dataset is the SNAP Email-Eu-core benchmark network (https://snap.stanford.edu/data/email-Eu-core.html) [1, 2]. Full 64-character SHA-256 cryptographic checksums for both edge lists and department labels were computed immediately upon acquisition to establish a local integrity baseline for automated pipeline verification and reproducibility, permanently documented in the accompanying standalone analysis notebook and repository verification suite.

Preprocessing Pipeline: Dual graph representations were constructed: G_{raw} retains all 1,005 nodes, 25,571 directed edges, and 642 self-loops. G_{analysis} excludes all 642 self-loops, preserving all 1,005 nodes and 24,929 interpersonal directed edges; 19 singleton nodes with only self-loops are preserved as degree-0 nodes with their official ground-truth department labels. The unweighted undirected projection $G_{\text{undirected}}$ preserves all 1,005 nodes and contains 16,064 undirected edges (8,865 reciprocal pairs, 71.12% reciprocity).

Software and Computing Environment: Executed under Python 3.10.6 on standard commodity hardware (operable within 8 GB RAM). Pinned versions in requirements.txt: NetworkX 3.4.2, SciPy 1.15.3, scikit-learn 1.5.2, pandas 2.3.3, NumPy 1.26.4, matplotlib 3.10.9, seaborn 0.13.2.

Algorithmic Parameters & Random Seeds: PageRank was computed using damping factor $d = 0.85$, uniform personalization, NetworkX dangling mass redistribution, and convergence tolerance 10^{-6} [3]. Louvain modularity optimization was performed with resolution $\gamma = 1.0$ and fixed random seed 42 [4, 5]. Deterministic tie-breaking is enforced by sorting by metric descending, then node ID ascending.

Execution Instructions: The accompanying analysis.ipynb notebook is completely self-contained and imports no local src/ modules. It automatically downloads the raw dataset from SNAP if absent, verifies cryptographic checksums, executes all analyses, and reproduces all figures and numerical tables from a fresh kernel in under 30 seconds.

References

Note: Per official DSA5101 instructions, the reference list is excluded from the eight-page limit. Appendices are not included.

[1] H. Yin, A. R. Benson, J. Leskovec, and D. F. Gleich, 'Local higher-order graph clustering,' in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 555–564.

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